

Matam Santosh

Software Developer

+918951955539 ◇ matamsantosh178@gmail.com ◇ Bangalore , Karnataka, India ◇ [LinkedIn](#)

SUMMARY

C++ Developer with over 2+ years of experience in embedded systems and automotive software development. Proficient in C++, Python, and embedded software design, with expertise in CAN bus communication, real-time control, and cloud-integrated software solutions. Experienced in working with ISO 26262 standards and adept in hardware-in-the-loop (HIL) testing and on-board software integration.

EXPERIENCE

Software Developer

Sep '22 — Present

Cognizant

Bengaluru, India

- Developed C++ and Python code to manage real-time control features, including LED and steering wheel operations, for autonomous vehicle systems.
- Developed an automated testing framework in C++ to support unit and integration testing for complex software modules.
- Implemented multithreading and concurrency control to enhance application performance.
- Designed and implemented communication protocols (CAN, TCP/IP) to optimize vehicle-to-vehicle data transfer, improving latency by 20%.
- Created and validated vehicle test scripts using an in-house Python conversion tool and HIL simulations to ensure safety and compliance with ISO 26262 standards.
- Led the development of OTA systems, managing version control, rollback mechanisms, and secure updates to vehicle software and firmware.
- Collaborated with cross-functional teams on AUTOSAR integration, MCAN implementations, and testing of embedded modules.

EDUCATION

B.Tech in Electronics and Communication Engineering, REVA UNIVERSITY (GPA: 9.32)

Aug '18 — Jul '22

Bengaluru, India

SKILLS

Programming Languages C++, Python, Embedded C

Operating system Linux , RTOS, Ubuntu, Windows

Automotive Protocols CAN bus, LIN, TCP/IP, MQTT

Tools Git, Visual studio , Cmake, GCC, GDB

CERTIFICATIONS

[Google Cloud Certified Associate Cloud Engineer](#)

Google

Jan '23

[Microsoft Certified Azure Data Fundamentals](#)

Microsoft

Feb '23

PROJECTS

SDV Accelerator, Cognizant

Dec '23 — Present

Bangalore , India

- **Client:** - Cognizant
Description: - As part of SDV initiative in cloud native development Software Defined Vehicle (SDV) technology, that would put their own Cloud infrastructure at the heart of the next generation mobility concept their partner with selected Automotive OEMs to incubate and mature the overall concept, while the OEMs plot their transition towards the Centralized Compute Automotive platforms, being the physical hardware to host the on-board part of the SDV solution.
Technology used: -AWS, OTA, Ubuntu, C++,Python,docker.
Role Played:-
 - Developed and maintained high-performance C++ applications for real-time data processing.
 - Developed a C++ code for communication between two boards.
 - OTA end to end workflow
 - Full automated process workflow for Cloud Native development and virtualization.
 - Porting of the stacks on the multiple target hardware platforms like NXP- Toradex(i.MX 8QM)

- **Client:** - Stellantis

Description : This Test-Automation Project deals with automation of vehicle features. Simulating the Vehicle Functionalities by developing the Test scripts and validating the scripts. Analysing the Vehicle function requirement related to Body, Chassis, ADAS Features & develop the

Test script for Vehicle functions using the conversion tool (Python based) To make sure the script stability, we do acceptance validation in the system HIL bench. System HIL having BCM, RFHM, ETM, IPC, DCSD, etc.,

Technology used : Python, CAN, UDS, Automation Tool Kit, Control Desk, Dspace Automotion Desk

Role Played :

- Developing a Test script from Test plan with the help of Vehicle Functionalities (VF's), CAN/LIN Database, Model Interface sheet.
- Doing ADP generation for the developed Test Scripts in Automation Tool Kit (ATK). i.e. to convert the excel Format test script (.xlsx) to Automation Desk project (.adpx)
- Working on Hil Execution for the developed Test Script in Automation Desk.
- To fix the Test Script issues in trail Executions.
- To generate HTML Report from Automation Desk project (ADP) in Automation Tool Kit (ATK) after the completion of Execution.
- To generate Test Report in excel format with the help of ATK Tool.